Name_____ Date_____ Period_____

LESSON 4-5 · HOMEWORK PACKET · LEVEL 1 · SHEET 1 OF 2

Find the Whole

Maryland Standard 6.AT.4 · Learning target – I can find the whole when I know a part and the percent it represents.

REMEMBER

- The **part** is what you know. The **whole** is 100%.
- Step **down** to a friendly percent, then build **up** to 100%.
- Percent below 100%? The whole is **greater** than the part.

WORKED EXAMPLE

24 is 40% of what number?

Step 1 40% is 24, so 10% is 24 ÷ 4 = 6.

Step 2 100% is 10 × 6 = 60.

Step 3 Check: 0.4 × 60 = 24, so it checks.

Answer **60**

WORD BANK

part (parte) – the amount you know

whole (el entero) – the full amount; it is 100%

percent (por ciento) – out of 100

double number line (doble recta numérica) – two lines that match amounts and percents

PRACTICE

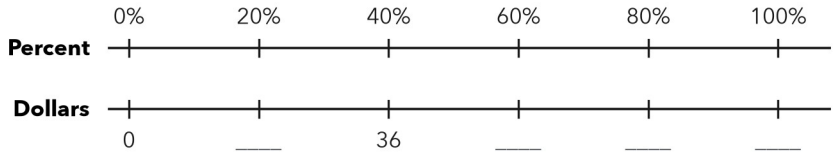
show your work · label units

1. Bike Helmet. Andre has saved $36. That is 40% of the cost of a bike helmet. Fill in the double number line. Then find the cost of the helmet.

Step 1 40% ÷ 2 = 20%. So $36 ÷ 2 = the 20% amount.

Step 2 Add that amount at each jump.

Step 3 The amount under 100% is the cost.



Cost_____ dollars

2. Find each whole.

Step 1 Find 10% (or 5% or 25%) first.

Step 2 Multiply up to 100%.

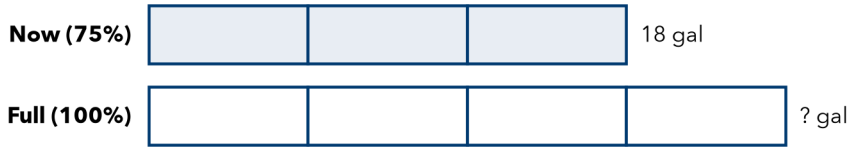
a. 13 is 10% of _____ **b.** 11 is 25% of _____ **c.** 21 is 30% of _____

d. 32 is 40% of _____ **e.** 7 is 5% of _____ **f.** 54 is 60% of _____

3. Water Tank. A water tank is 75% full. It holds 18 gallons now. Each part of the tape diagram is 25% of the full tank. How many gallons does the tank hold when it is full?

Step 1 3 parts = 18 gallons. Find 1 part.

Step 2 The full tank is 4 parts.



Answer_____ gallons

4. Chapter Book. Lena has read 120 pages. That is **75%** of her book. How many pages are in the **whole** book?

Step 1 75% is 3 parts of 25%. Find 1 part: 120 ÷ 3.

Step 2 Multiply by 4 to get 100%.

Answer_____ pages

5. Reason. 20 is 40% of number A. 20 is also 80% of number B. Without solving, which number is greater, A or B? Explain. Then find both numbers.

Frame ___ *is greater because 20 is a* ___ *percent of it. A =* ___ *and B =* ___.

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #____

AT HOME Tell someone at home: if 10 is 50% of a number, what is the number? Explain how you know. **Sentence frame:** "I know the number is ___ because ___."

Name_____ Date_____ Period_____

LESSON 4-5 · HOMEWORK PACKET · LEVEL 1 · SHEET 2 OF 2

Solve for the Whole

Maryland Standard 6.AT.4 · Learning target – I can write and solve an equation to find the whole from a part and a percent.

REMEMBER

- Write **part = percent × whole**, with *w* for the whole.
- Solve by dividing: *w* = part ÷ decimal.
- **Check:** decimal × whole should give the part.

WORKED EXAMPLE

$18 is 60% of a shirt's first price. Find the first price.

Step 1 Write the equation: 0.6 × *w* = 18

Step 2 Divide: *w* = 18 ÷ 0.6 = 180 ÷ 6 = 30

Step 3 Check: 0.6 × 30 = 18, so it checks.

Answer **$30**

WORD BANK

variable (variable) – a letter, like *w*, for the whole you are looking for

equation (ecuación) – a math sentence with =

divide (dividir) – undo multiplying

check (comprobar) – multiply back to get the part

PRACTICE

show your work · label units

1. Solve each equation for *w*.

Step 1 Divide the number on the right by the decimal.

Step 2 Check: decimal × *w* gives the number.

a. 0.2 × *w* = 14 ____ **b.** 0.25 × *w* = 12 ____ **c.** 0.8 × *w* = 64 ____

2. 33 is 55% of what number? Write an equation and solve it.

Step 1 Change 55% to a decimal.

Step 2 Use decimal × *w* = part. The part is the number you know.

Step 3 Divide the part by the decimal.

a. Equation: _____

b. *w* = _____

3. Field Trip. A class has $315. That is **75%** of the trip cost. Find the total cost. How much **more** does the class need?

Step 1 Write $0.75 \times w = 315$ and divide.

Step 2 Subtract $315 from the total.

Show your work.

Total and still needed_____

4. Savings Goal. Tasha has saved $42. That is 35% of her goal. Write an equation and find her goal.

Frame *Equation:* ___ × *w* = ___. *Goal:* *w* = ___ ÷ ___ = $___.

Step 1 Change 35% to a decimal.

Step 2 Write decimal × *w* = the part you know.

Step 3 Divide the part by the decimal.

Equation and goal_____

5. Find the Error. Jamal solved "28 is 35% of *w*" as $w = 28 \div 35 = 0.8$. Explain his mistake. Solve and check.

Frame *Jamal's mistake is* ___. *35%* = ___, *so w = 28 ÷* ___ = ___.

TEST PRACTICE

choose the best answer

6. 45 is 30% of a number *w*. Which equation can you use to find *w*?

Step 1 Find the part (the number you know) and the decimal for 30%.

Step 2 The whole *w* is the number you do not know. Put it next to the decimal.

A $0.3 \times 45 = w$ **B** $0.3 \times w = 45$ **C** $45 \times w = 30$ **D** $w \div 45 = 0.3$

7. A team won 18 games. That is 60% of its games. How many games did it play?

Step 1 60% = 0.6. Write $0.6 \times w = 18$.

Step 2 Divide: $18 \div 0.6$.

A 10.8 **B** 24 **C** 30 **D** 78

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____

AT HOME Explain to someone at home why the whole must be bigger than the part when the percent is less than 100%.